

Lab 12 Template

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Problem 1

a)

First system is for non-linear pendulum function, second is for linear

```
v = theta'  
[theta; v]' = [v; -g/L.*sin(theta)]
```

```
v = theta'  
[theta; v]' = [v; -g/L.*theta]
```

b)

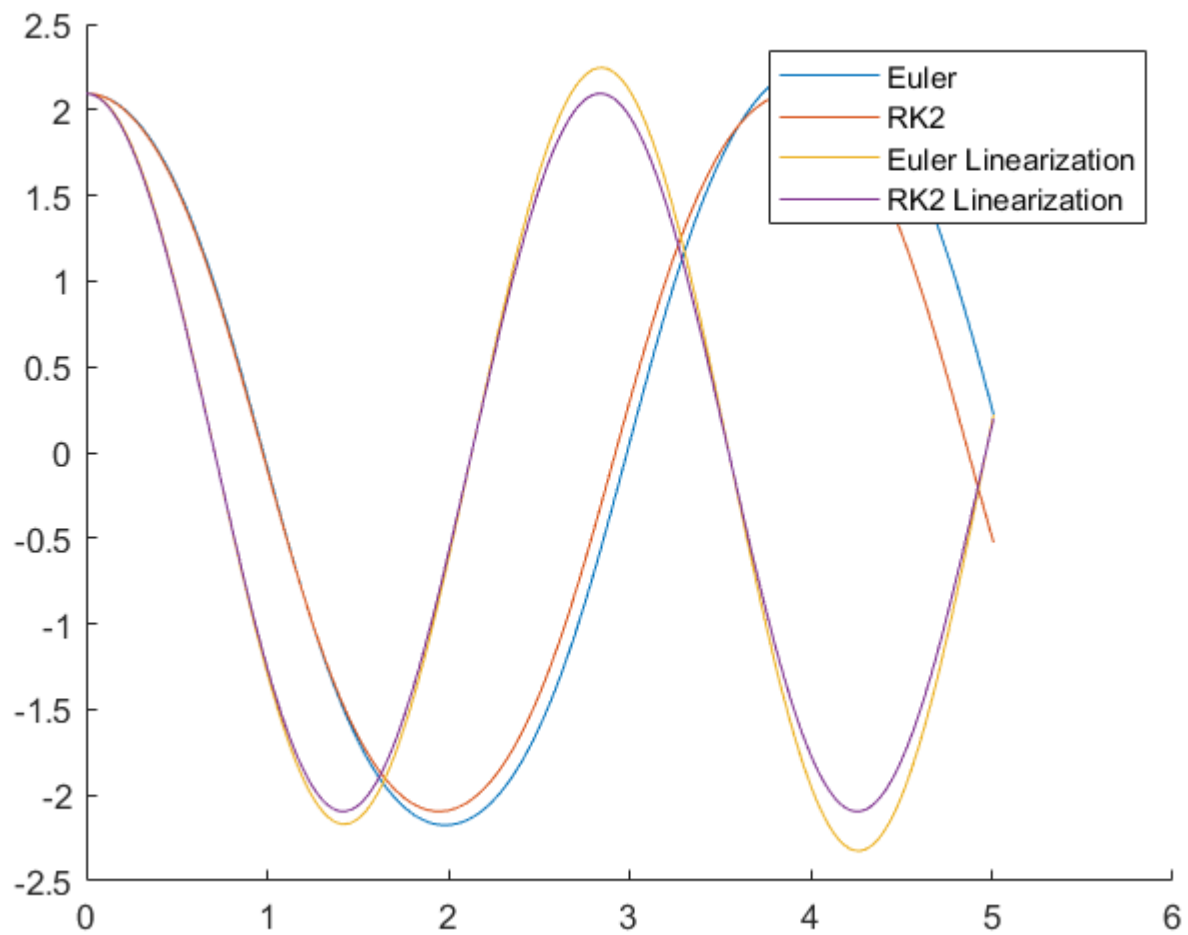
```
function [result] = MyVectorEulerMethod(f,t0,zn,tn,h)  
    i = t0;  
    while i <= tn  
        zn = [zn, zn(1:2,end) + h.*f(zn(1:2,end))];  
        i = i + h;  
    end
```

```
    result = zn;  
end
```

```
function [result] = MyVectorRK2(f,t0,zn,tn,h)  
    i = t0;  
    while i <= tn  
        v1 = f(zn(1:2,end));  
        v2 = f(zn(1:2,end)+h.*v1);  
        zn = [zn, zn(1:2,end) + (h/2).*(v1+v2)];  
        i = i + h;  
    end
```

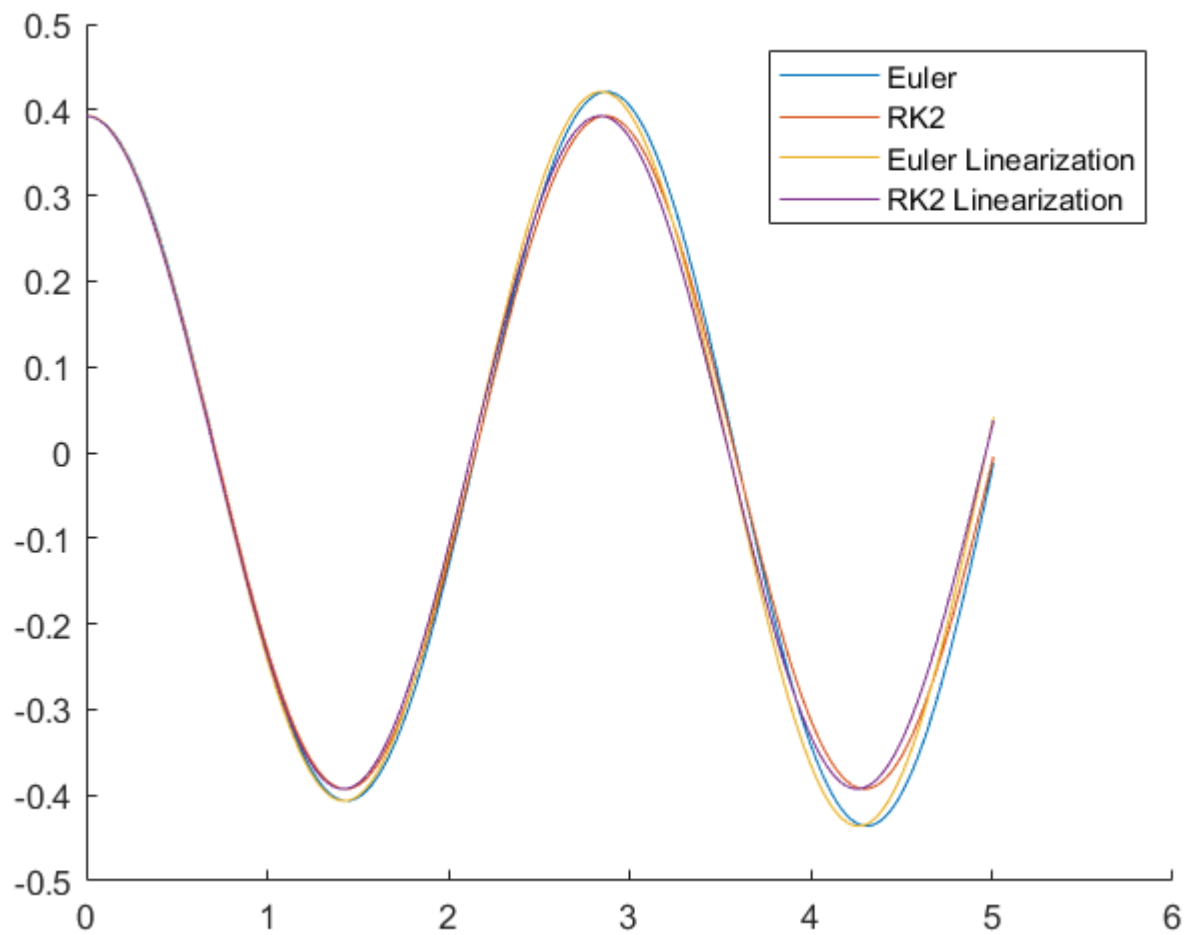
```
    result = zn;  
end
```

c)



It's pretty clear that RK2 does much better than Euler, as Euler seems to be increasing over time as seen by the higher amplitudes. This would suggest an increase in energy to the system, which is not what is occurring in an actual pendulum's motion. The linearization is also just very very very off. It's not a good approximation with this starting theta since the error is very high

d)



Here, the linearization works much better since we have a smaller starting theta value.

Problem 2

(a)

$$\begin{aligned} v &= y' \\ [y; v]' &= [v; -9v+8y] \end{aligned}$$

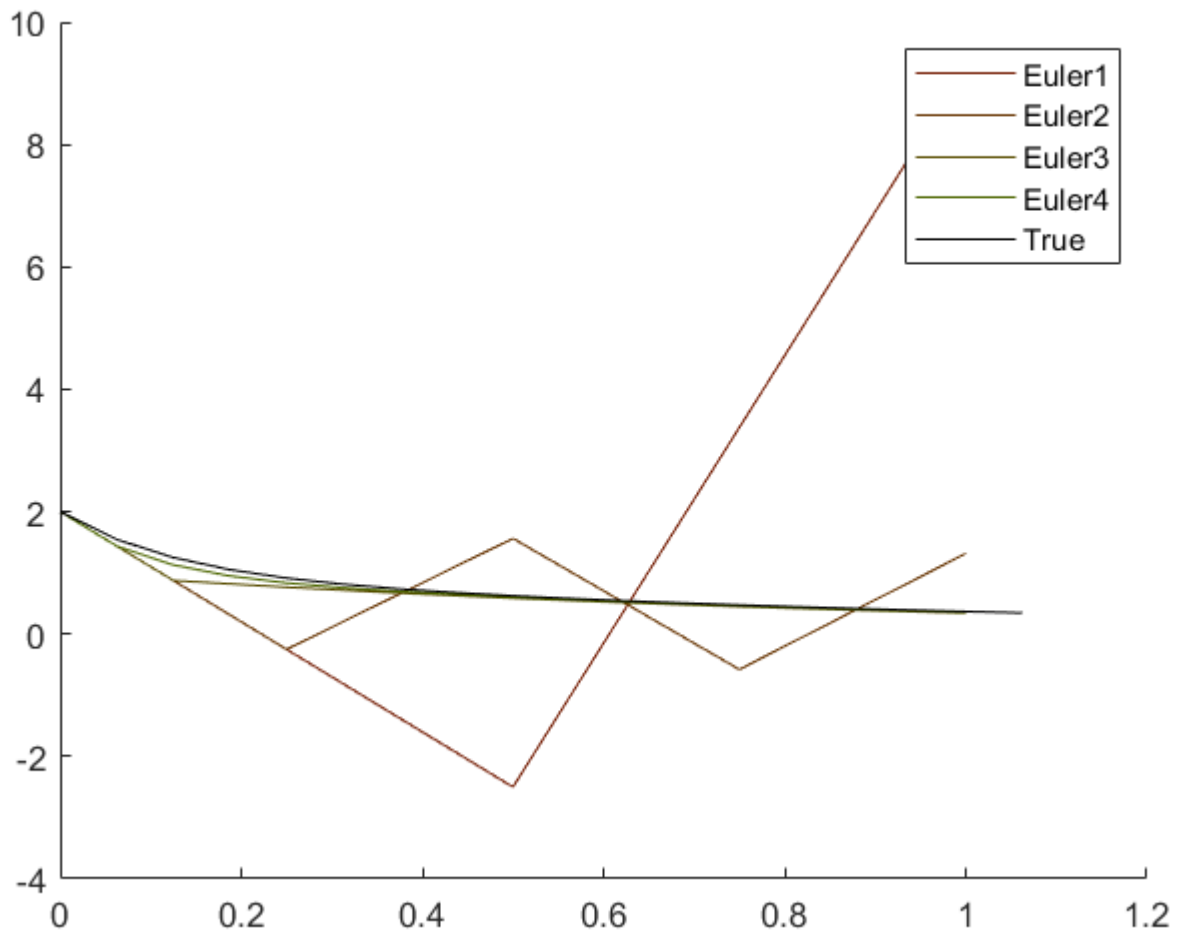
(b)

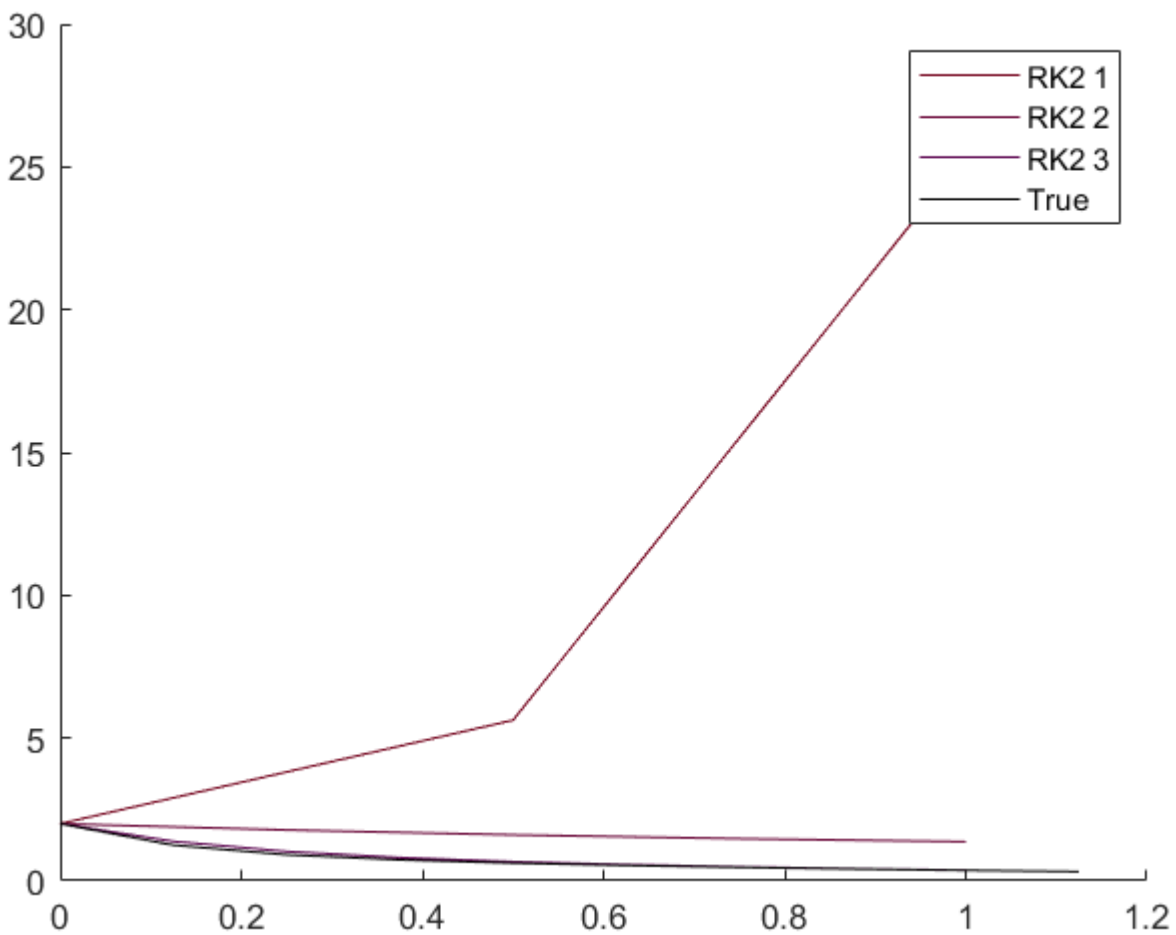
10x5 table

i	ErrEuler	ErrEulerRatio	ErrRK2	ErrRK2Ratio
1	8.8818	0	25.022	0
2	1.1641	0.13107	1.0043	0.040137
3	0.37538	0.32245	0.13244	0.13187
4	0.12147	0.3236	0.022821	0.17232
5	0.053232	0.43823	0.004668	0.20455

6	0.025141	0.47229	0.0010583	0.22672
7	0.012238	0.48679	0.00025223	0.23832
8	0.0060418	0.49368	6.1584e-05	0.24416
9	0.003002	0.49687	1.5216e-05	0.24708
10	0.0014963	0.49844	3.7818e-06	0.24854

(c)





As can be seen, RK2 took one less iteration in order for it to begin to match the end behaviour. This makes sense, as RK2 has faster convergence than Euler's, and so is much more accurate even at a higher step size. However, at the smallest step size, RK2 has less stability it seems and diverges much faster than Euler's. Since the step size is so large, what's likely occurring is that the Taylor error term (which is a constant times a power of h) is larger in RK2 than Euler for that specific h value.